SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club Submission
Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name AMONGX

Selected Domain DOMAIN 2

Selected SDG Track (1–17) SDG 8

Project / Solution Name Orbyte-hub

Name (Leader) SOH ZHENG HAO

Email & Phone (Leader) 0129281791

University Affiliation(s) HELP UNIVERSITY & Universiti Malaya

Number of Team Members 5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved. ***[Maximum 250 words]***

Malaysian micro, small and medium enterprises (MSMEs) face increasing difficulty keeping up with changing government policies and digital compliance requirements. Requirements such as e-Invoice, tax-related policies, employment regulations and other SME-related guidelines can change over time, while the information is often distributed across different official sources and lengthy documents.

This creates a practical problem for SME owners and their staff. Many small businesses do not have dedicated compliance teams or the resources to continuously monitor government updates, interpret complex requirements and determine whether a particular change affects their business. As a result, important policy changes may be overlooked, misunderstood or handled too late.

The problem is especially significant because MSMEs form a large part of Malaysia's business ecosystem. When compliance processes rely heavily on manual checking, SMEs may spend valuable time searching documents, checking invoices and determining what actions are required. This increases administrative workload and creates a risk of inconsistent or incorrect compliance decisions.

Therefore, SMEs need a simpler way to understand relevant policies, identify changes that may affect their business, and take the appropriate action. we addresses this gap by bringing policy intelligence and compliance workflows into one SME-focused platform, helping business owners move from simply finding information to understanding what it means for their business and what they should do next.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Orbyte InvoisReady is a compliance support platform designed to help Malaysian SMEs handle e-Invoice and changing business-policy requirements with less manual work. The system converts unstructured inputs such as WhatsApp messages, receipts and PDF documents into structured information that can be checked against defined compliance rules.

AI is used selectively for tasks that require interpretation of unstructured information. It extracts relevant fields such as transaction amounts, buyer information and tax-related details, and can assist in interpreting newly published policy documents. AI outputs are treated as proposed information rather than final decisions.

Compliance decisions remain deterministic. Python-based rules perform threshold checks, eligibility validation and date-based logic so that the same input produces consistent results. Important actions also require human review and approval, while audit records provide traceability for decisions.

The system uses a web frontend, Python/Flask backend, structured policy data, database storage and GenAI API integration. Skills from the SDC workshops in GenAI API integration, database architecture and cloud deployment will be applied to build and deploy a functional prototype.

The overall principle is: AI reads, rules decide, and humans approve.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

No. Our project did not utilize any official SDC sponsor-provided GenAI API or Cloud Platform.

The project uses JavaScript and TypeScript for the frontend and backend application logic, with SQL and Supabase PostgreSQL for database storage and management. Google Gemini is integrated to support AI-based processing and understanding of unstructured text and documents. ChatGPT is used as an AI-assisted development tool for coding, debugging and system design, while Replit is used as the development and deployment environment. The compliance workflow uses structured rules implemented in JavaScript and TypeScript to perform deterministic validation, threshold checks and eligibility decisions. Database records are used to store and manage company information, invoices, policy information and application data.

The overall architecture connects the web interface to the TypeScript/JavaScript backend, which communicates with the Supabase database and Gemini AI services. Inputs such as invoices, receipts, WhatsApp messages and policy information are processed through the appropriate AI and rule-based workflows before producing compliance results, alerts or recommended actions.

The project did not utilize any official SDC sponsor-provided GenAI API or Cloud Platform. Google Gemini, ChatGPT and Replit were used through their respective services for AI functionality, development assistance and deployment.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Orbyte Hub was designed based on the practical needs of Malaysian SMEs. Instead of requiring users to read lengthy regulations, the platform presents important information in a simpler format and helps users identify what needs attention and what action should be taken.

The design process followed four stages: understanding the problem, designing the workflow, developing the prototype, and testing. The team identified common difficulties SMEs face when handling e-Invoice information and keeping track of policy changes. Based on these needs, two main workflows were developed: InvoisReady for invoice and receipt compliance checking, and Policy Monitoring & Policy Alerts for tracking relevant regulatory changes.

A functional web prototype was developed with different access levels for staff, owners, and administrators. Testing uses realistic SME scenarios, including invoice threshold checks, policy updates, access restrictions, and audit records.

For deployment, the frontend and backend will be hosted as a web application, with Supabase/PostgreSQL used for persistent data storage. Scheduled workflows will support policy-source synchronisation, while environment variables will protect API keys and configuration settings.

The system will be further improved through testing and feedback, with focus on reliability, usability, data consistency, and accurate compliance workflows

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess [Innovation and Creativity as well as Feasibility of build]. Vague or unsupported projections may weaken your case for advancing to Phase 2.

Metric Category	**Your Response**
Target SDG Sub-Target(s)	Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises.

Metric Category	Your Response
Identify the exact UN SDG subtarget your project addresses *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Malaysian MSMEs represent a very large share of the business ecosystem, creating a significant need for accessible digital tools that reduce administrative and compliance workload. The project's prototype baseline will also measure the average time required to manually process and assess selected SME compliance scenarios
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Average processing time per compliance case, measured from receiving an invoice/document/policy input to producing a clear compliance result and recommended action.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	At least 30% reduction in average processing time for the selected prototype scenarios compared with the team's manual baseline, while maintaining a traceable rule and evidence path for each compliance decision

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

yes All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

yes The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

yes The proposed solution targets a functional, deployed application rather than a theoretical business case study.

All fields in this template have been completed within the word limits stated.

yes

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

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7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature] 

[Date] 25/10/2026